Write a program to simulate deadlock detection.

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#include <stdio.h>
#define MAX_PROCESSES 10
#define MAX_RESOURCES 10

int allocation[MAX_PROCESSES][MAX_RESOURCES];
int request[MAX_PROCESSES][MAX_RESOURCES];
int available[MAX_RESOURCES];
int resources[MAX_RESOURCES];
int work[MAX_RESOURCES];
int marked[MAX_PROCESSES];

int main() {
 int num_processes, num_resources;

 printf("Enter the number of processes: ");
 scanf("%d", &num_processes);

 printf("Enter the number of resources: ");
 scanf("%d", &num_resources);

 // Input total resources
 for (int i = 0; i < num_resources; i++) {
 printf("Enter the total amount of Resource R%d: ", i + 1);
 scanf("%d", &resources[i]);
 }

 // Input request matrix
 printf("Enter the request matrix:\n");
 for (int i = 0; i < num_processes; i++) {
 for (int j = 0; j < num_resources; j++) {
 scanf("%d", &request[i][j]);
 }
 }

 // User Input allocation matrix
 printf("Enter the allocation matrix:\n");
```

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for (int i = 0; i < num_processes; i++) {
for (int j = 0; j < num_resources; j++) {
scanf("%d", &allocation[i][j]);
}
}

// Initialization of the available resources
for (int j = 0; j < num_resources; j++) {
available[j] = resources[j];
for (int i = 0; i < num_processes; i++) {
available[j] -= allocation[i][j];
}
}

// Mark processes with zero allocation
for (int i = 0; i < num_processes; i++) {
int count = 0;
for (int j = 0; j < num_resources; j++) {
if (allocation[i][j] == 0) {
count++;
} else {
break;
}
}
if (count == num_resources) {
marked[i] = 1;
}
}

// Initialize work with available
for (int j = 0; j < num_resources; j++) {
work[j] = available[j];
}

// Mark processes with requests <= work
for (int i = 0; i < num_processes; i++) {
int can_be_processed = 1;
if (marked[i] != 1) {
```

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    for (int j = 0; j < num_resources; j++) {
    if (request[i][j] > work[j]) {
    can_be_processed = 0;
    break;
    }
    }
    if (can_be_processed) {
    marked[i] = 1;
    for (int j = 0; j < num_resources; j++) {
    work[j] += allocation[i][j];
    }
    }
    }
    }

    // Check for unmarked processes (deadlock)
    int deadlock = 0;
    for (int i = 0; i < num_processes; i++) {
    if (marked[i] != 1) {
    deadlock = 1;
    break;
    }
    }

    if (deadlock) {
    printf("Deadlock detected\n");
    } else {
    printf("No deadlock possible\n");
    }

    return 0;
   }
```

Output:

Enter the number of processes: 3
Enter the number of resources: 3
Enter the total amount of Resource R1: 4
Enter the total amount of Resource R2: 5
Enter the total amount of Resource R3: 7

Enter the request matrix:
1 2 3
4 5 6
7 8 9
Enter the allocation matrix:
5 6 7 1 2 3 7 8 9
Deadlock detected